

Lecturer: dr hab. inż. Bogdan Trawiński, prof. uczelni

IT Project Management Support

Lecture 1

Project Planing

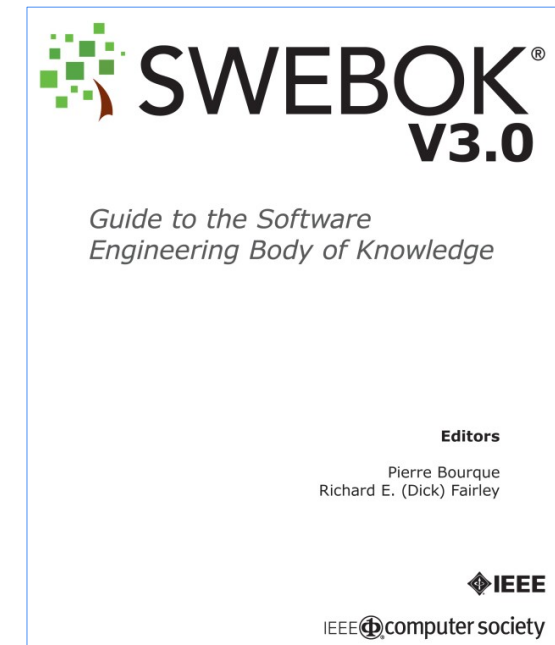
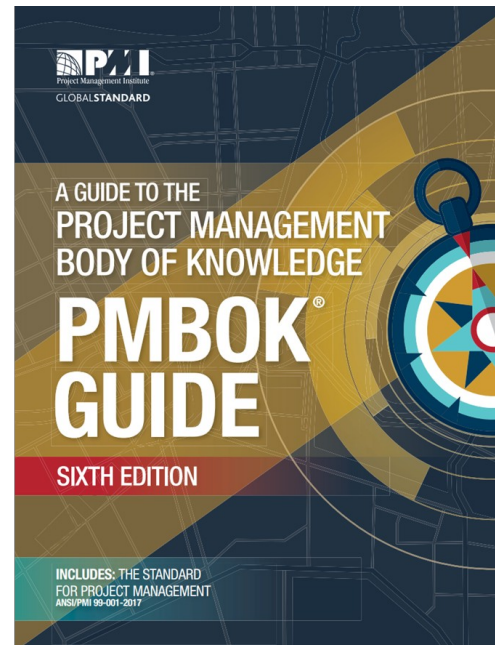
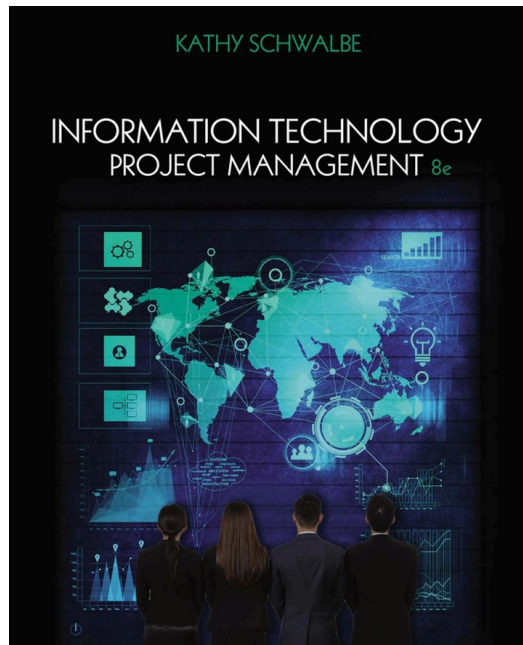
- Introduction
- Project Management basic notions
- Project Charter & RACI Matrix

Scope of the Lecture

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- Lecture 02 Project Management Software.pptx
- Lecture 03 Total Cost of Ownership.pptx
- Lecture 04 Requirement Management.pptx
- Lecture 05 Work Breakdown Structure.pptx
- Lecture 06 Project Scheduling.pptx
- Lecture 07 Earned Value Management.pptx
- Lecture 08 Managing Project Risk.pptx
- Lecture 09 Managing Teams (1).pptx
- Lecture 09 Managing Teams (2).pptx
- Lecture 11 Project Communications.pptx
- Lecture 12 Myers–Briggs personality types .pptx
- Lecture 13 Big Five personality traits.pptx
- Lecture 14 DISC personality traits.pptx

Literature

- Project Management Institute. A Guide to the Project Management Body of Knowledge (PMBOK Guide) – 6th Edition 2017
- Kathy Schwalbe: Information Technology Project Management, 8th Edition. Cengage Learning 2016
- SWEBOK - Guide to the Software Engineering Body of Knowledge v.3.0 IEEE 2014.
- **The Internet**





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Project Management Basic Notions

Definition of a Project

According to the PMBOK Guide:

A project is a temporary endeavor undertaken to create a unique product, service, or result.

In other words, a project

- is a sequence of tasks bounded by time, resources and required results;
- has a defined outcome and deliverables;
- has a deadline;
- has a budget limiting number of people, supplies and capital.

Project characteristics include:

- Accomplishment with shared resources often only available on part-time basis
- Cross-functional team work may be required
- Uncertainty and potential change during execution
- Changes to the way the business operates
- Specific deadlines, time and resource constraints.

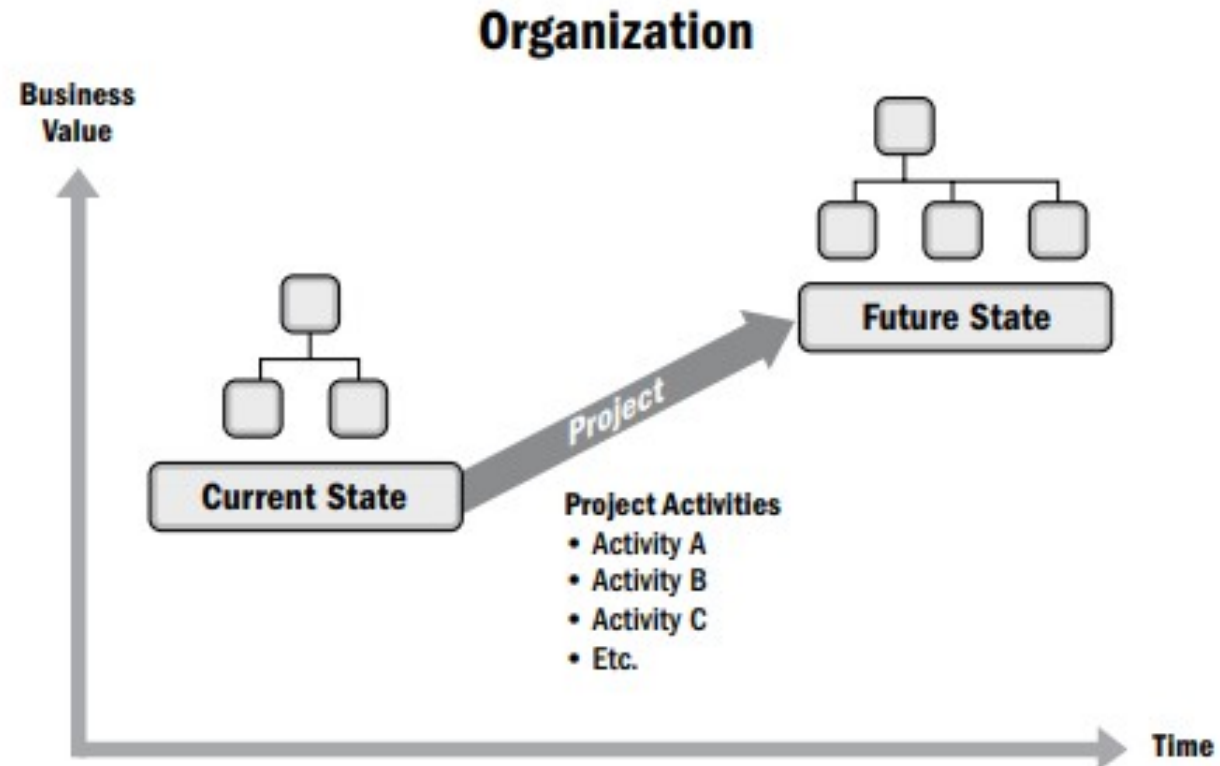
Definition of a Project Management

What does managing a project include?

According to the PMBOK Guide, managing a project includes:

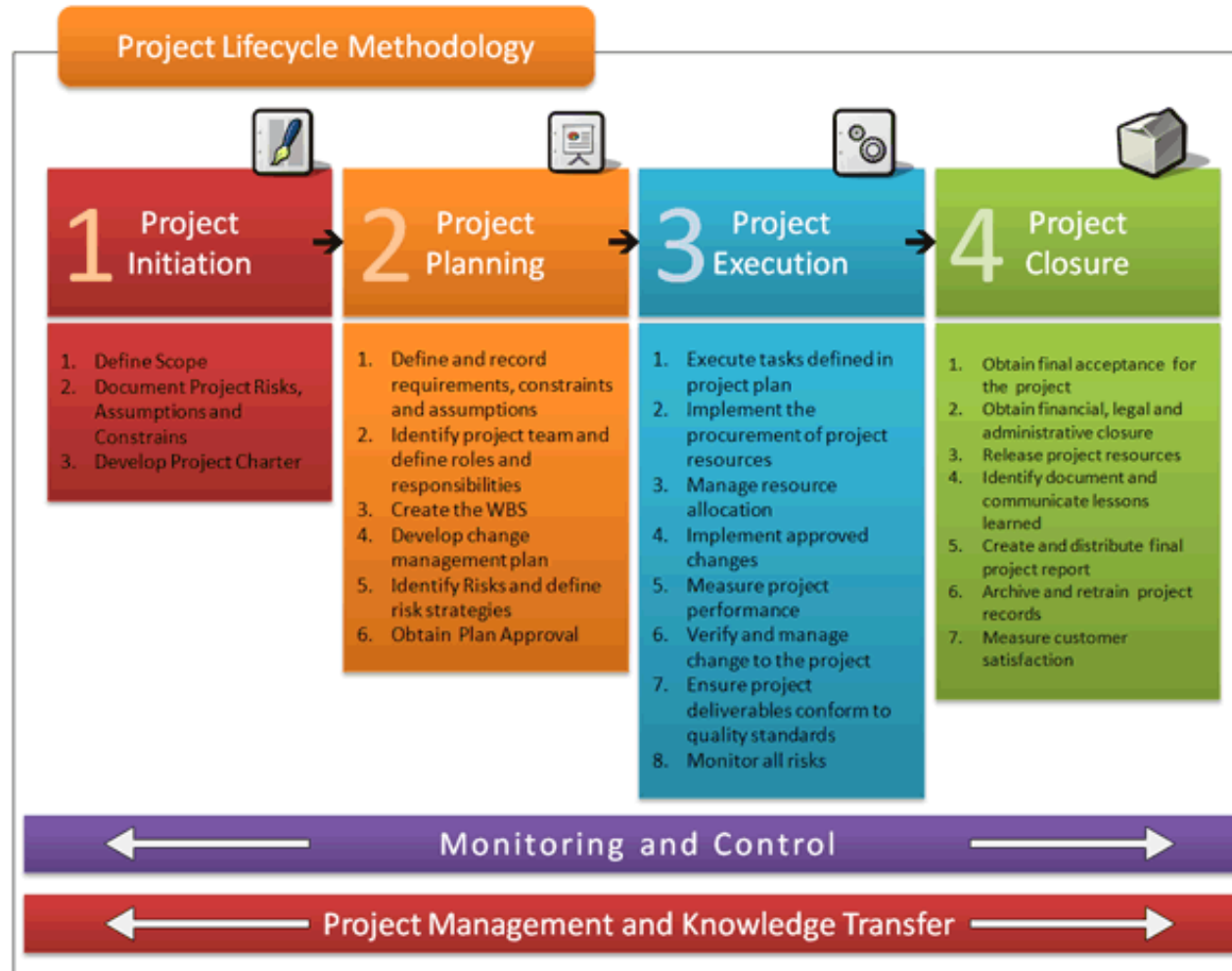
- Identifying the business issues and risks to be addressed by the project;
- Clarifying/Identifying requirements;
- Establishing clear and achievable objectives;
- Balancing competing demands for quality, scope, resources, time and cost;
- Adapting to different concerns and expectations of stakeholders.

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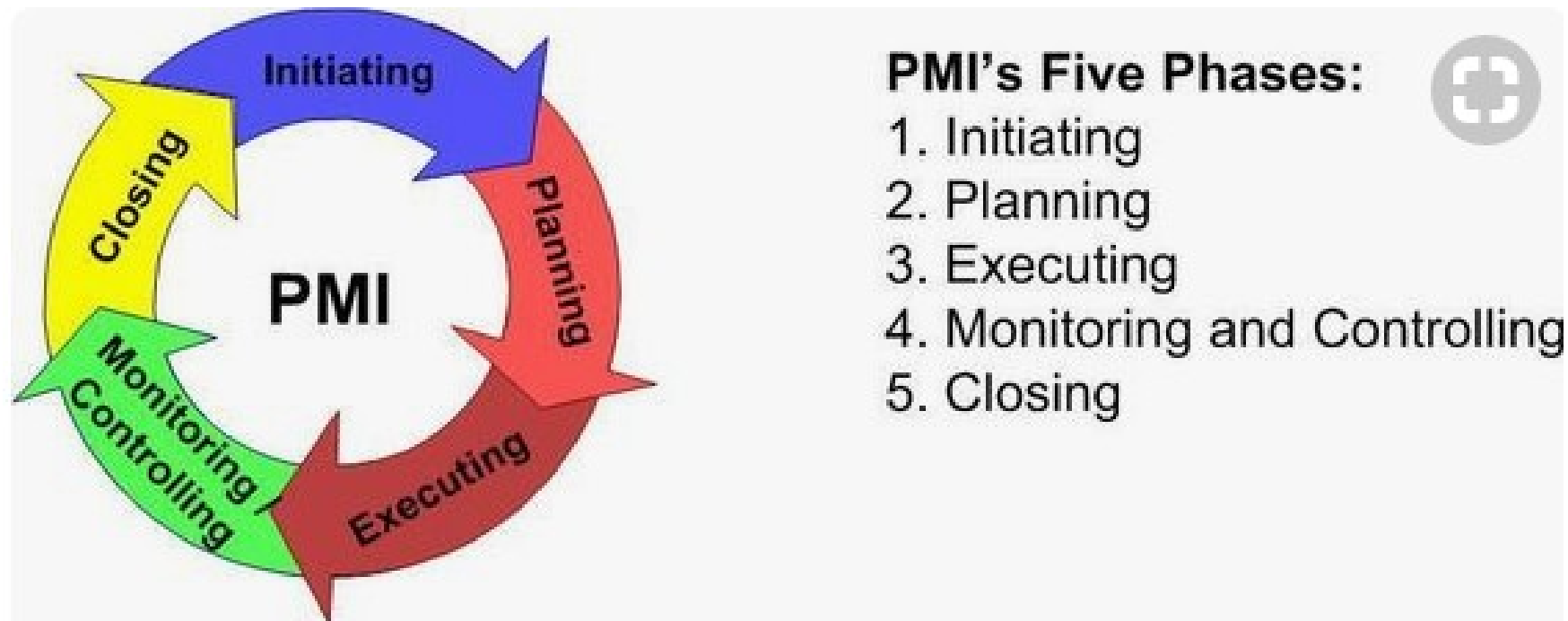
From a business perspective, a project is aimed at moving an organization from one state to another state in order to achieve a specific objective.

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Project Lifecycle

Stages of a Project according to PMI



Stages of a Project according to PMI

Projects are divided into six stages:

1. **Definition**
2. **Initiation**
3. **Planning**
4. **Execution**
5. **Monitoring & Control**
6. **Closure**

Each stage is characterised by a distinct set of activities that take the project from its first idea to its conclusion. Each stage is of equal importance and contributes to the overall success of the project.

1. Definition

Before a project starts the project manager must make sure the project goals, objectives, scope, risks, issues, budget, timescale and approach have been defined. This must be communicated to all the stakeholders to get their agreement. Any differences of opinion need to be resolved before work starts.

2. Initiation

This is perhaps the most important stage of any project as it sets the terms of reference within which the project will be run. If this is not done well, the project will have a high likelihood of failure. The initiation stage is where the business case is declared, scope of the project decided and stakeholder expectations set. Time spent on planning, refining the business case and communicating the expected benefits will help increase the likelihood of success. It is tempting to start working quickly, but a poor initiation stage often leads to problems and even failure.

3. Planning

The key to a successful project is in the planning. Creating a project plan is the first task you should do when undertaking any project. Often project planning is ignored in favour of getting on with the work. However, many people fail to realise the value of a project plan in saving time, money and many other problems.

4. Execution

Doing the work to deliver the product, service or wanted outcome. Most of the work related to the project is realised at this stage and needs complete attention from the project manager.

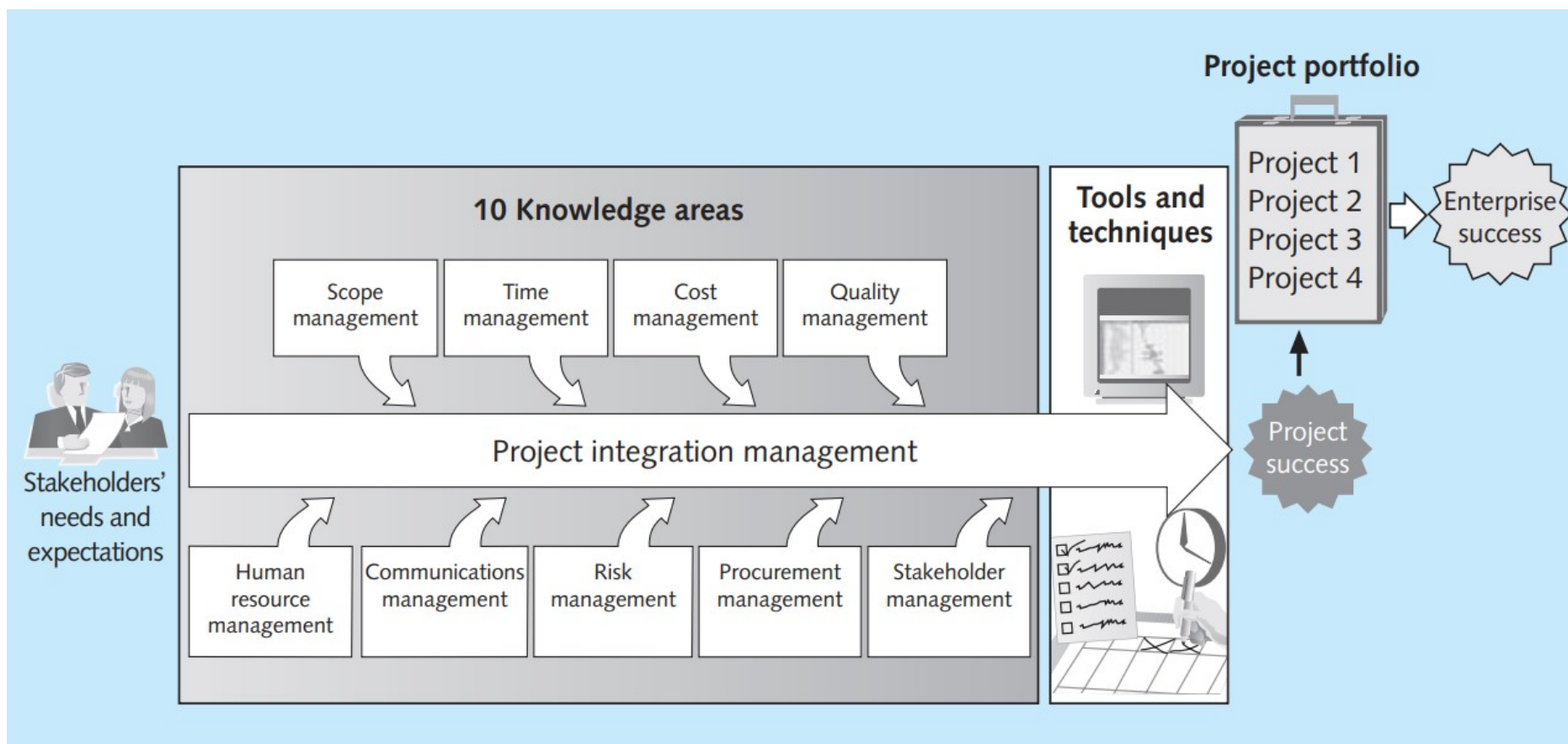
5. Monitoring & Control

Once the project is running it is important the project manager keeps control. This is achieved by regular reporting of issues, risks, progress and the constant checking of the business case to ensure that expected benefits will be delivered and are still valid. A project that is not controlled is out of control.

6. Closure

Often neglected, it is important to ensure that a project is closed properly. Many projects never end because there is no formal sign-off. It is important to get the customers agreement that a project has ended and that no more work will be carried out. Once closed, the project manager should review the project and record the good and bad points so successes can be repeated and failures avoided. A project that is not closed will continue to consume resources, albeit slowly.

10 Project Management Knowledge Areas



10 Project Management Knowledge Areas

1. **Project scope management** involves defining and managing all the work required to complete the project successfully.
2. **Project time management** includes estimating how long it will take to complete the work, developing an acceptable project schedule, and ensuring timely completion of the project.
3. **Project cost management** consists of preparing and managing the budget for the project.
4. **Project quality management** ensures that the project will satisfy the stated or implied needs for which it was undertaken.
5. **Project human resource management** is concerned with making effective use of the people involved with the project.

10 Project Management Knowledge Areas

6. **Project communications management** involves generating, collecting, disseminating, and storing project information.
7. **Project risk management** includes identifying, analyzing, and responding to risks related to the project.
8. **Project procurement management** involves acquiring or procuring goods and services for a project from outside the performing organization.
9. **Project stakeholder management** includes identifying and analyzing stakeholder needs while managing and controlling their engagement throughout the life of the project.
10. **Project integration management** is an overarching function that affects and is affected by all of the other knowledge areas.



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Project Charter & RACI Matrix

Project charter is a document issued by the project initiator or sponsor that formally authorizes the existence of a project and provides the project manager with the authority to apply organizational resources to project activities.

It documents the high-level information on the project and on the product, service, or result the project is intended to satisfy, such as:

- Project purpose;
- Measurable project objectives and related success criteria;
- High-level requirements;
- High-level project description, boundaries, and key deliverables;
- Overall project risk;
- Summary milestone schedule;
- Preapproved financial resources;
- Key stakeholder list;
- Project approval requirements (i.e., what constitutes project success, who decides the project is successful, and who signs off on the project);
- Project exit criteria (i.e., what are the conditions to be met in order to close or to cancel the project or phase);
- Assigned project manager, responsibility, and authority level; and
- Name and authority of the sponsor or other person(s) authorizing the project charter.

At a high level, the project charter ensures a common understanding by the stakeholders of the key deliverables, milestones, and the roles and responsibilities of everyone involved in the project.

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TABLE 4-1 Project charter for the next-gen DNA-sequencing instrument completion project

Project Title: Next-gen DNA-Sequencing Instrument Completion Project

Date of Authorization: February 1

Project Start Date: February 1

Projected Finish Date: November 1

Key Schedule Milestones:

- Complete first version of the software by June 1
- Complete production version of the software by November 1

Budget Information: The firm has allocated \$1.5 million for this project, and more funds are available if needed. The majority of costs for this project will be internal labor. All hardware will be outsourced.

Project Manager: Nick Carson, (650) 949-0707, ncarson@dnaconsulting.com

Project Objectives: The Next-gen DNA-sequencing instrument project has been under way for three years. It is a crucial project for our company. This is the first charter for the project; the objective is to complete the first version of the instrument software in four months and a production version in nine months.

Main Project Success Criteria: The software must meet all written specifications, be thoroughly tested, and be completed on time. The CEO will formally approve the project with advice from other key stakeholders.

Approach:

- Hire a technical replacement for Nick Carson and a part-time assistant as soon as possible.
- Within one month, develop a clear work breakdown structure, scope statement, and Gantt chart detailing the work required to complete the Next-gen DNA-sequencing instrument.
- Purchase all required hardware upgrades within two months.
- Hold weekly progress review meetings with the core project team and the sponsor.
- Conduct thorough software testing per the approved test plans.

Example of Project Charter
(simplified)

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ROLES AND RESPONSIBILITIES			
Name	Role	Position	Contact Information
Ahmed Abrams	Sponsor	CEO	aabrams@dnaconsulting.com
Nick Carson	Project Manager	Manager	ncarson@dnaconsulting.com
Susan Johnson	Team Member	DNA expert	sjohnson@dnaconsulting.com
Renyong Chi	Team Member	Testing expert	rchi@dnaconsulting.com
Erik Haus	Team Member	Programmer	ehaus@dnaconsulting.com
Bill Strom	Team Member	Programmer	bstrom@dnaconsulting.com
Maggie Elliot	Team Member	Programmer	melliot@dnaconsulting.com
Sign-off: (Signatures of all the above stakeholders)			
<i>Ahmed Abrams</i>		<i>Nick Carson</i>	
<i>Susan Johnson</i>		<i>Renyong Chi</i>	
<i>Erik Haus</i>		<i>Bill Strom</i>	
<i>Maggie Elliot</i>			
Comments: (Handwritten or typed comments from above stakeholders, if applicable)			
<i>“I want to be heavily involved in this project. It is crucial to our company’s success, and I expect everyone to help make it succeed.”—ahmed abrams</i>			
<i>“The software test plans are complete and well documented. If anyone has questions, do not hesitate to contact me.”—Renyong Chi</i>			

Example of Project Charter
(simplified)



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 INTERNATIONAL UNIVERSITY - HCMC School of Computer Science & Engineering		Project Charter
Prepared by:	Date:	
	Version:	
Project Name:		
Sponsor:		
Project Summary:		
Project Manager:	Beneficiary:	
Other stakeholders:		
PM Responsibilities:		

Template of Project Charter
(simplified)

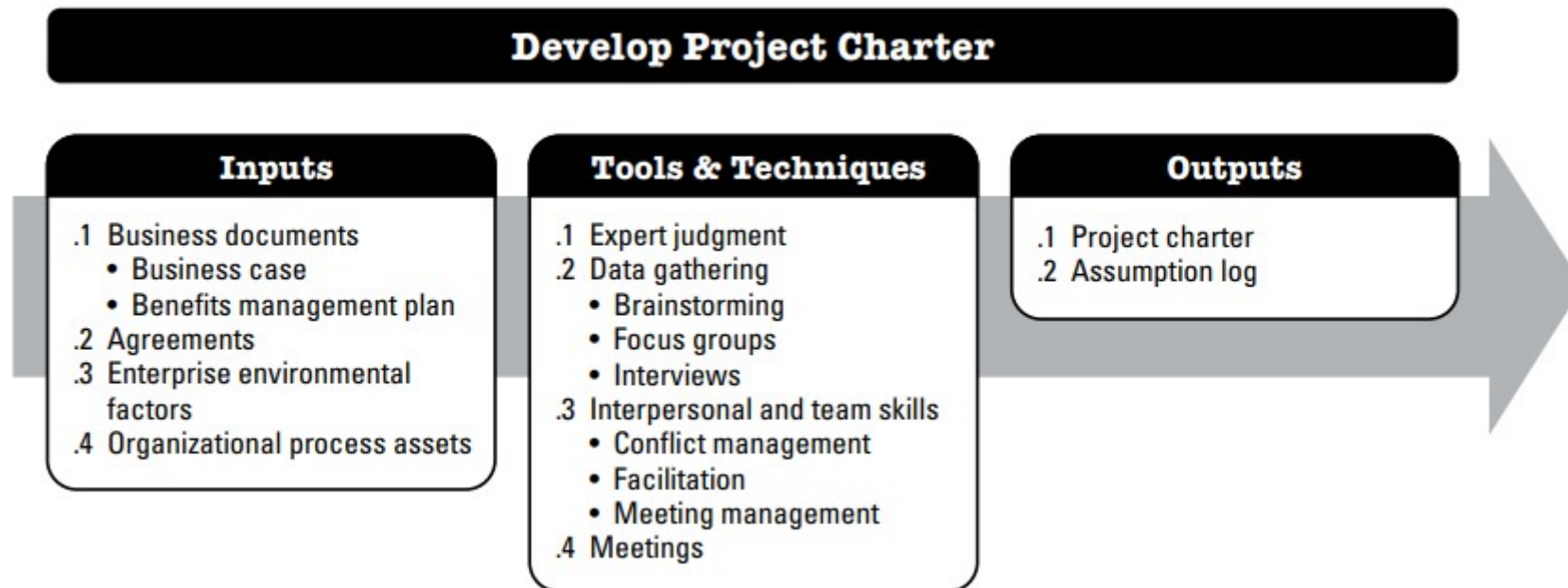
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Objectives:	
Scope:	
Milestone schedule:	
Budget:	
Risks:	
Assumptions:	
Sponsor sign off:	PM sign off:

Template of Project Charter
(simplified)

Develop Project Charter is the process of developing a document that formally authorizes the existence of a project and provides the project manager with the authority to apply organizational resources to project activities.

The key benefits of this process are that it provides a direct link between the project and the strategic objectives of the organization, creates a formal record of the project, and shows the organizational commitment to the project.



RACI matrix

RACI matrix is a responsibility assignment matrix that describes the participation by various roles in completing tasks or deliverables for a project or business process.

The RACI model is a straightforward tool used for identifying roles and responsibilities and avoiding confusion over those roles and responsibilities during a project.

The acronym RACI stands for:

Responsible: The person who does the work to achieve the task. They have responsibility for getting the work done or decision made. As a rule this is one person; examples might be a business analyst, application developer or technical architect.

Accountable: The person who is accountable for the correct and thorough completion of the task. This must be one person and is often the project executive or project sponsor. This is the role that responsible is accountable to and approves their work.

Consulted: The people who provide information for the project and with whom there is two-way communication. This is usually several people, often subject matter experts (SME).

Informed: The people kept informed of progress and with whom there is one-way communication. These are people that are affected by the outcome of the tasks, so need to be kept up-to-date.



RACI Definitions

R

- Who is Responsible
- The person who is assigned to do the work

A

- Who is Accountable
- The person who makes the final decision and has the ultimate ownership

C

- Who is Consulted
- The person who must be consulted before a decision or action is taken

I

- Who is Informed
- The person who must be informed that a decision or action has been taken

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RACI Matrix						
Task \ Role	Role 1	Role 2	Role 3	Role 4	Role 5	Role 6
Task 1	R		C			
Task 2	R		A		C	
Task 3	R	R	I	A		I
Task 4			C		R	
Task 5	A	C		I		
Task 6		I	C	C		R

Responsible

Person assigned to do the activity.

Accountable

Person makes final decisions and has the ownership.

Consulted

Person who must be consulted before a decision or action is taken.

Informed

Person who must be informed when a decision or action has been taken.

Template of RACI Matrix (simplified)

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RACI Matrix

[Project Title]

Roles and Responsibilities

Responsible, Accountable, Consulted, Informed

Deliverable or Task	Status	ROLES					Project Manager	Technical Lead	Name or Role	Name or Role	Name or Role	Consultant	Name or Role	Name or Role	Name or Role	Name or Role
		Sponsor	Name or Role	Name or Role	Name or Role	Name or Role										
		Sponsor / Leadership					Project Team					Other Resources				
Phase 1																
Deliverable/Task 1		A	R				I									
Deliverable/Task 2		A		R			I									
Phase 2																
Deliverable/Task 1		C	I				A	R								
Deliverable/Task 2			I				A		R							
Phase 3																
Deliverable/Task 1			I				A	I		R		C				
Deliverable/Task 2			I				A	I	R			C				
Phase 4																
Deliverable/Task 1				I			A	R					C			
Deliverable/Task 2				I			A		R							
Insert new rows above this one																

D	Driver	Assists those who are responsible for a task.
R	Responsible	Assigned to complete the task or deliverable.
A	Accountable	Has final decision-making authority and accountability for completion. Only 1 per task.
S	Support	Provides support during implementation.
C	Consulted	An adviser, stakeholder, or subject matter expert who is consulted before a decision or action.
I	Informed	Must be informed after a decision or action.

Template of RACI Matrix (simplified)

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RACI Matrix Template																			
ROLE Project Deliverable (or Activity)		Executive Sponsor Project Sponsor Steering Committee Advisory Committee Role #5				Project Manager Tech Lead Functional Lead SME				Project Team Member Developer Administrative Support Business Analyst Role #4 Role #5				Consultant PMO Role #3 Role #4					
		Project Leadership				Project Team Members				Project Sub-Teams				External Resources					
Initiate Phase Activities																			
Request Review by PMO		A/C	R/A				R/A	A/C		C									
Submit Project Request							R										A		
Research Solution		I					R/A	A/C	A/C	C				C			C		
Develop Business Case		I	A/C	I	I		R/A	C	C	C				C			C	C	
Plan Phase Activities																			
Create Project Charter		C	C				R/A	C	C	C				C			C		
Create Schedule		I	I	I	I		R/A	C	C	C	C	C	C	C			C	I	
Create Additional Plans as Required		I	I	I			R/A				I	I	I	I			C	I	
Execute Phase Activities																			
Build Deliverables		C/I	C/I	C/I	C/I			R/A	R/A	R/A	R/A	R/A					A/C		
Create Status Report		I	I	I	I		R/A	R/A	R/A	R/A							C	I	
Control Phase Activities																			
Perform Change Management				C	C	C		R	A	A	A						C	I	
Close Phase Activities																			
Create Lessons Learned		C	C	C	C		R/A	C	C	C	C	C	C	C			C	C	
Create Project Closure Report		I	I	I	I		R/A	I	I	I	I	I	I	I				I	

SME - Subject-Matter Expert - is a person who is an authority in a particular area or topic.

PMO - Project Management Office is a department within an enterprise that defines and maintains standards for project management within the organization.

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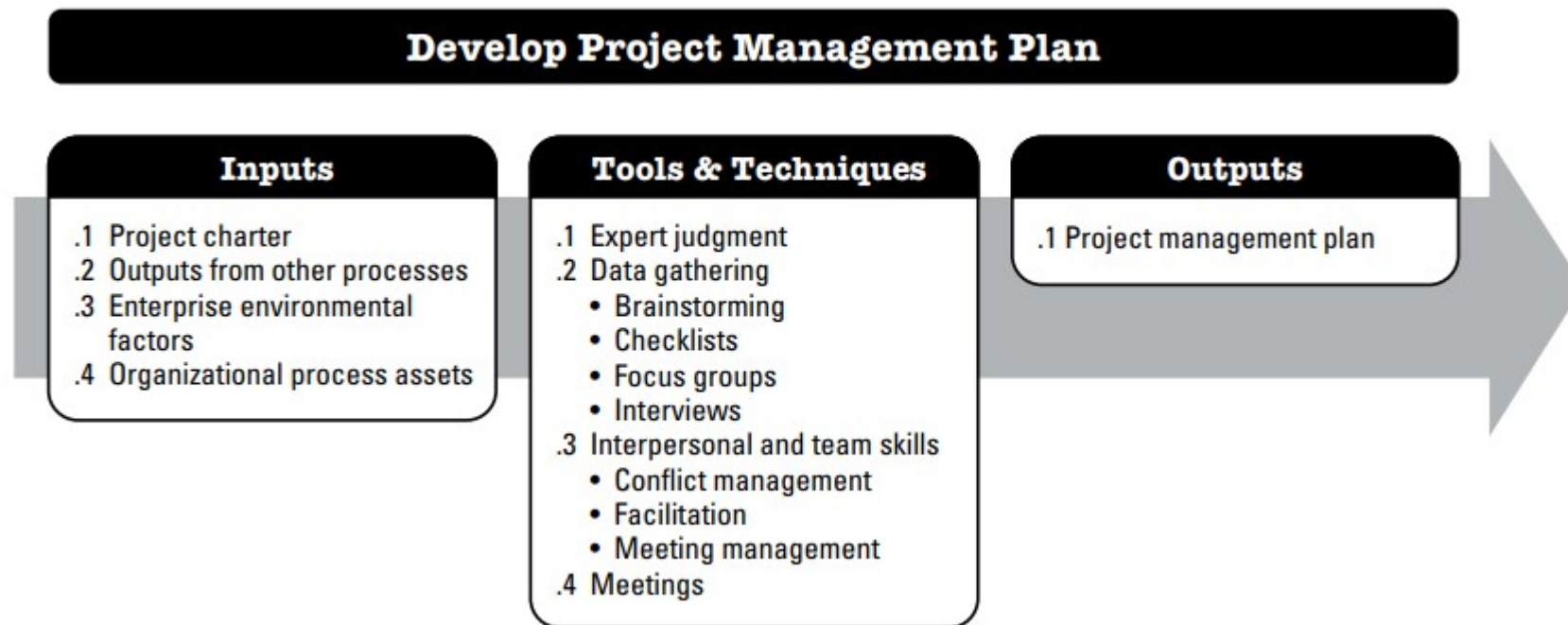
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Creating a RACI Matrix (step-by-step)

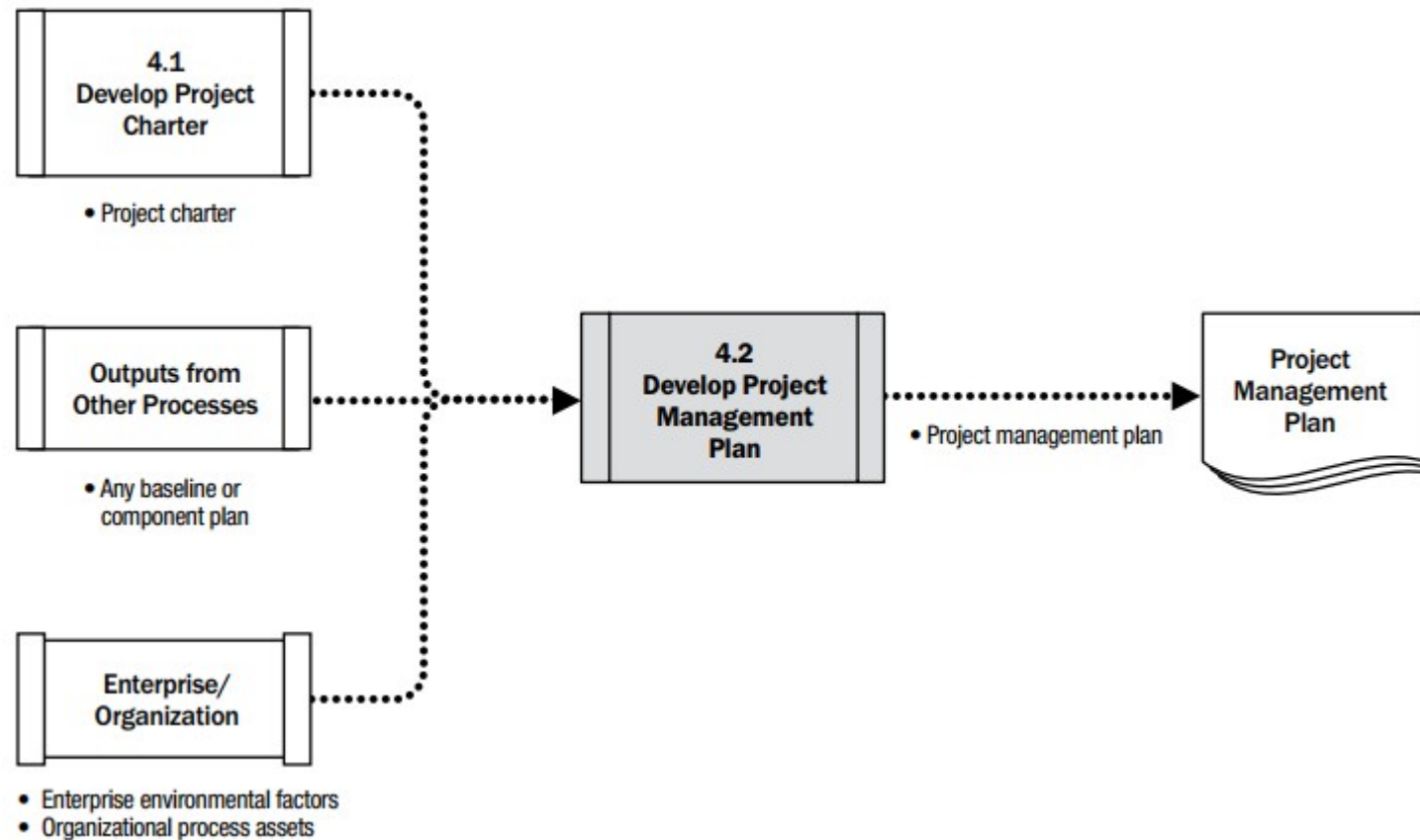
1. Identify all the tasks involved in delivering the project and list them on the left-hand side of the chart in completion order.
2. Identify all the project roles and list them along the top of the chart.
3. Complete the cells of the chart identifying who has the responsibility, the accountability and who will be consulted and informed for each task.
4. Ensure every task has a role responsible and a role accountable for it.
5. No tasks should have more than one role accountable. Resolve any conflicts where there is more than one for a particular task.
6. Share, discuss and agree on the RACI Matrix with your stakeholders before your project starts.

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Develop Project Management Plan is the process of defining, preparing, and coordinating all plan components and consolidating them into an integrated project management plan. The key benefit of this process is the production of a comprehensive document that defines the basis of all project work and how the work will be performed.



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Develop Project Management Plan: Data Flow Diagram

Project Management Plan and Project Documents

Project Management Plan	Project Documents	
1. Scope management plan	1. Activity attributes	19. Quality control measurements
2. Requirements management plan	2. Activity list	20. Quality metrics
3. Schedule management plan	3. Assumption log	21. Quality report
4. Cost management plan	4. Basis of estimates	22. Requirements documentation
5. Quality management plan	5. Change log	23. Requirements traceability matrix
6. Resource management plan	6. Cost estimates	24. Resource breakdown structure
7. Communications management plan	7. Cost forecasts	25. Resource calendars
8. Risk management plan	8. Duration estimates	26. Resource requirements
9. Procurement management plan	9. Issue log	27. Risk register
10. Stakeholder engagement plan	10. Lessons learned register	28. Risk report
11. Change management plan	11. Milestone list	29. Schedule data
12. Configuration management plan	12. Physical resource assignments	30. Schedule forecasts
13. Scope baseline	13. Project calendars	31. Stakeholder register
14. Schedule baseline	14. Project communications	32. Team charter
15. Cost baseline	15. Project schedule	33. Test and evaluation documents
16. Performance measurement baseline	16. Project schedule network diagram	
17. Project life cycle description	17. Project scope statement	
18. Development approach	18. Project team assignments	



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Thank You very much
for your attention